

MENTAL HEALTH DATA ANALYTICS

PROJECT DOCUMENTATION

Strategic Planning & Implementation Roadmap

Data Source: SAMHSA (samhsa.gov/data)

Prepared per: Project Documentation Guidelines

Team Information

Name	Role
Mohammed Mahmoud Abdelrazek	Project Lead & Data Architect
Shaza Yousef	BI Analyst (NSDUH)
Habiba Mohammed	BI Analyst (MH-CLD)
Khadija Osama	BI Analyst (TEDS-A)
Mohannad Mohammed	AI & Data Engineering Associate

Documentation Deliverables Schedule

Per the Project Documentation Guidelines, all documents are uploaded to GitHub against the following deadlines:

Item	Student Deadline	Graduate Deadline
Project Planning & Management	2/20/2026	1/27/2026
Requirements Gathering	4/20/2026	3/1/2026
System Analysis & Design	5/1/2026	4/3/2026
Implementation (Source Code & Execution)	7/10/2026	5/17/2026
Final Presentation & Testing & Reports	7/17/2026	5/24/2026

1. Project Planning & Management

1.1 Project Proposal

The project builds a modern, optimized data pipeline covering preprocessing, compression, exploratory analysis, and interactive visualization of SAMHSA mental health trends. It aims to process and prepare data with high-performance engineering using Parquet compression, identify prevalence and patterns across demographic groups, analyze treatment admission and discharge outcomes, build a robust Star Schema for macro-level dataset connectivity, and deliver an interactive Power BI dashboard with a semantic layer.

Objectives

- **Process & Prepare** – High-performance engineering with Parquet compression.
- **Identify Trends** – Prevalence and patterns across demographic groups.
- **Analyze Care** – Treatment admission and discharge outcome patterns.
- **Robust Modeling** – Star Schema for macro-level dataset connection.
- **Interactive Delivery** – Optimized Power BI dashboard with semantic layers.

Scope & Data Sources

Included SAMHSA datasets:

- **NSDUH** – National Survey on Drug Use and Health.
- **TEDS-A** – Treatment Episode Data Set: Admissions.
- **MH-CLD** – Mental Health Client-Level Data.

Excluded sources:

- **DAWN & URS** – Raw data not public.
- **N-SUMHSS** – Facility-level data only; contains no patient-level records.
- **TEDS-D** – Corrupted / unsolvable data; files on website cannot be processed.

1.2 Project Plan (Timeline, Milestones & Deliverables)

The project is organized into eight phases spanning January to July 2026:

Phase	Dates	Deliverables
Phase 0–1	Jan – May 1, 2026	Planning & Requirements: scope & role assignment, stakeholder identification, architecture documentation
Phase 2–3	May 2 – Jun 6, 2026	Acquisition & Preprocessing: ingestion of 3 SAMHSA datasets, Parquet compression, missing value handling
Phase 4	Jun 7 – Jun 27, 2026	Data Modeling & EDA: star schema implementation, semantic layer/views, trend analysis & EDA
Phase 5	Jun 28 – Jul 2, 2026	Power BI Development: dataset loading, interactive filter setup, AI Q&A integration
Phase 6	Jul 2 – Jul 8, 2026	Review & Documentation: team quality review, README & code upload, final data validation

Phase 7	Jul 8 – Jul 17, 2026	Final Presentation: deck preparation, team rehearsal, dashboard walkthrough
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1.3 Task Assignment & Roles

Lead & Architect — Mohammed Mahmoud

- Overseeing the entire project lifecycle and team coordination.
- Architecting the cloud infrastructure (AWS S3 & Amazon Athena).

BI Analysts — Shaza, Habiba, Khadija

- Execute preprocessing scripts (select columns, handle nulls).
- Prepare decoding logic from SAMHSA codebooks.
- Conduct Exploratory Data Analysis (EDA) on assigned SAMHSA datasets.
- Develop Power BI Star Schema models and complex DAX measures.
- Manage the Semantic Layer for unified reporting.
- Design and integrate interactive visualization dashboards.

BI Analysts / AI Bot — Mohannad

- Developing the core AI Bot (Text-to-SQL integration logic).

AI & Data Engineering Associate

- Collaborating on the AI Bot architecture and development.
- Optimizing Athena SQL queries for performance.
- Assisting in automating data pipeline components.
- Leading the project documentation and preparing the final presentation.

1.4 Risk Assessment & Mitigation Plan

Risk	Mitigation
File Size Limits	Local feature selection and 80% compression via Parquet conversion.
Linkage Issues	Use macro-level dimensions (State/Year) instead of Patient IDs.
Calculation Errors	Early conversion of 9-coded flags to Nulls in the pipeline.
Performance	On-the-fly decoding via Semantic Layers and DAX measures.
Team Coordination	Weekly check-ins and a centralized GitHub repository for version control.

1.5 KPIs (Key Performance Indicators)

Category	KPI
Engineering	KPI 1 — 80% storage reduction via Parquet conversion.
Engineering	KPI 2 — 100% conversion of 9-coded flags to Null.
Engineering	KPI 3 — Unified naming/types across annual files.
Analysis	KPI 4 — Star schema with 0 direct many-to-many relations.

Analysis	KPI 5 — Minimum 5 documented findings per dataset.
Delivery	KPI 6 — Minimum 5 interactive dashboard pages.
Delivery	KPI 7 — 6 out of 7 milestones completed on time.
Delivery	KPI 8 — Final submission by July 10, 2026.

2. Requirements Gathering

Stakeholder Analysis

Stakeholder needs and expectations identified for the project:

Stakeholder	Needs	Expectations
Project Lead & Data Architect	Ensure project delivery, cloud infrastructure stability, and team coordination	Timely completion, high-quality dashboards, scalable architecture
BI Analysts	Clean, decoded, and well-modeled datasets for analysis and reporting	Reliable data, reusable semantic models, accurate DAX calculations
AI & Data Engineering Associate	Structured datasets and optimized Athena queries for AI integration	Fast query execution, standardized schema, successful Text-to-SQL responses
SAMHSA	Proper use of publicly available datasets while preserving data integrity	Accurate interpretation of variables and responsible data usage
Decision Makers (Healthcare Managers & Policy Makers)	Clear insights into mental health, substance use, and treatment trends	Easy-to-understand dashboards that support evidence-based decisions
Researchers & Analysts	Ability to explore relationships between demographics, depression, substance use, and treatment	Interactive filtering, drill-down capabilities, downloadable insights
External Reviewers / Instructors	Well-documented project demonstrating data engineering and BI best practices	Clear documentation, reproducible workflow, professional presentation

User Stories & Use Cases

The dashboard delivered by this project covers Substance Abuse Treatment Admissions (TEDS-A, 2021–2023). The following user stories and use cases reflect that dataset and dashboard:

User Story 1

As a **healthcare policymaker**, I want to *analyze treatment admissions by demographic characteristics* so that *I can identify the most affected population groups*.

Use Case:

1. User opens the dashboard.
2. Selects Year, Gender, Age Group, and Education Level filters.
3. Dashboard updates KPIs and charts instantly.
4. User identifies the demographic groups with the highest treatment admissions.

User Story 2

As a **public health analyst**, I want to *analyze primary substance use across different demographic groups* so that *I can identify substance abuse patterns and trends*.

Use Case:

1. User filters Primary Substance, Gender, Age Group, and State.
2. Dashboard displays the distribution of treatment admissions.
3. User compares the most commonly reported substances among different groups.
4. User exports insights for reporting.

User Story 3

As a **healthcare executive**, I want to *view summarized KPIs* so that *I can monitor treatment admission trends without reviewing raw datasets*.

Use Case:

1. User views KPI cards.
2. Reviews Total Cases, Most Common Primary Substance, Total Primary Substances, Total Routes, and Average Waiting Days.
3. Uses the dashboard during strategic meetings and decision-making sessions.

User Story 4

As a **healthcare planner**, I want to *analyze treatment admissions across states* so that *I can allocate healthcare resources more effectively*.

Use Case:

1. User opens the Geographic Analysis page.
2. Selects Year and Primary Substance filters.
3. Dashboard highlights states with the highest treatment admissions.
4. User identifies high-demand regions for healthcare planning.

User Story 5

As a **treatment program manager**, I want to *analyze waiting times by employment status and demographic characteristics* so that *I can improve access to treatment services*.

Use Case:

1. User filters Employment Status, Gender, and Age Group.
2. Dashboard displays average waiting days.
3. User identifies the groups experiencing the longest waiting times.
4. User recommends improvements to reduce treatment delays.

User Story 6

As a **mental health specialist**, I want to *analyze psychiatric diagnoses alongside treatment admissions* so that *I can better understand patients' mental health needs*.

Use Case:

1. User filters Diagnosis, Primary Substance, and Year.
2. Dashboard displays treatment admissions by diagnosis.
3. User identifies the most common psychiatric conditions among patients.
4. User exports the analysis for clinical reporting.

Additional Use Cases

- Compare treatment admissions across years.
- Analyze treatment admissions by state.

- Explore demographic distributions.
- Compare male and female treatment admissions.
- Analyze primary substance trends.
- Analyze treatment services.
- Analyze waiting time by employment status.
- Compare diagnoses across demographic groups.
- Identify the most affected states.
- Export dashboard reports.
- Apply multiple slicers simultaneously.
- Navigate between dashboard pages using buttons.
- View detailed drill-through reports.

Functional Requirements

- Display interactive KPI cards.
- Analyze treatment admissions by year.
- Analyze demographic characteristics.
- Analyze primary substances.
- Analyze treatment services.
- Analyze geographic distribution by state.
- Analyze waiting days.
- Analyze diagnosis and mental health indicators.
- Provide interactive filtering across all pages.
- Enable drill-through and cross-filtering between visuals.
- Allow exporting dashboard insights.

Non-functional Requirements

- Fast dashboard performance.
- User-friendly interface.
- Interactive visualizations.
- Responsive filtering.
- Accurate and consistent data.
- Secure data access.
- Easy navigation between dashboard pages.

3. System Analysis & Design

3.1 Problem Statement & Objectives

SAMHSA publishes large, fragmented mental health datasets (NSDUH, TEDS-A, MH-CLD) that are difficult to compress, link, and analyze at scale. The project addresses this by building an optimized ingestion and modeling pipeline, and delivering the resulting insights through an interactive Power BI dashboard with an integrated AI Q&A layer.

Use Case Diagram & Descriptions

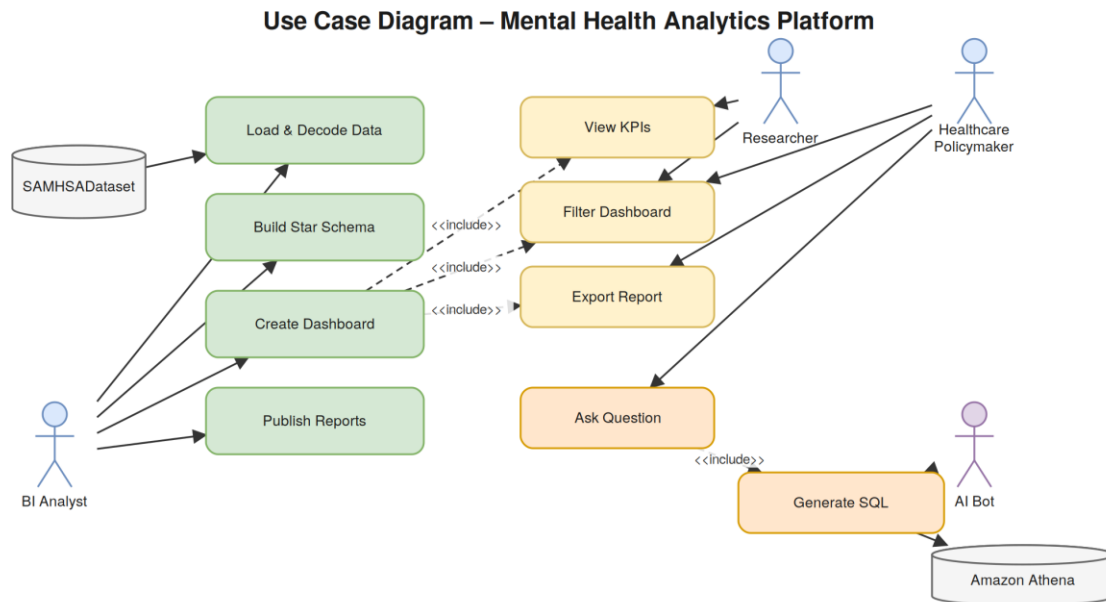


Figure 2 — Use Case Diagram: actors (Healthcare Policymaker, BI Analyst, Researcher, AI Bot) and their interactions with the data pipeline and Power BI dashboard.

Actors & Descriptions

- SAMHSA Dataset — external data source providing raw NSDUH, TEDS-A, and MH-CLD files.
- BI Analyst — loads and decodes data, builds the Star Schema, creates and publishes dashboards.
- Researcher — views KPIs and applies filters to explore trends across demographic groups.
- Healthcare Policymaker — filters, exports reports, and asks natural-language questions via the AI Bot.
- AI Bot — receives natural-language questions and generates SQL executed against Amazon Athena.

Functional & Non-Functional Requirements

Full lists are detailed in Section 2 (Requirements Gathering). In summary, the system must ingest, decode, and compress three SAMHSA datasets into a Star Schema with a Power BI semantic layer and an AI Text-to-SQL Q&A bot (functional), while meeting the project's KPI-driven targets for storage reduction, data quality, modeling integrity, and on-time delivery (non-functional).

Software Architecture

The architecture centers on a cloud data lake and query engine feeding a semantic-layer BI model:

- Storage: AWS S3 (raw and Parquet-compressed datasets).

- Query engine: Amazon Athena (SQL over S3, optimized for performance).
- Modeling layer: Star Schema with Dim_YEAR and Dim_State dimensions.
- Presentation layer: Power BI with DAX measures and a semantic layer.
- AI layer: Text-to-SQL AI Bot for natural-language Q&A over Athena.

3.2 Database Design & Data Modeling

ER Diagram (Entity-Relationship Diagram)

The data model follows a Star Schema. Three fact-like decoded views — `mh_cld_decoded_view` (MH-CLD), `teds_a_decoded_view` (TEDS-A), and `nsduh_decoded_view` (NSDUH) — connect to two shared dimension tables, `Dim_YEAR` and `Dim_State`, rather than to each other directly, avoiding many-to-many relationships between datasets (per KPI 4).

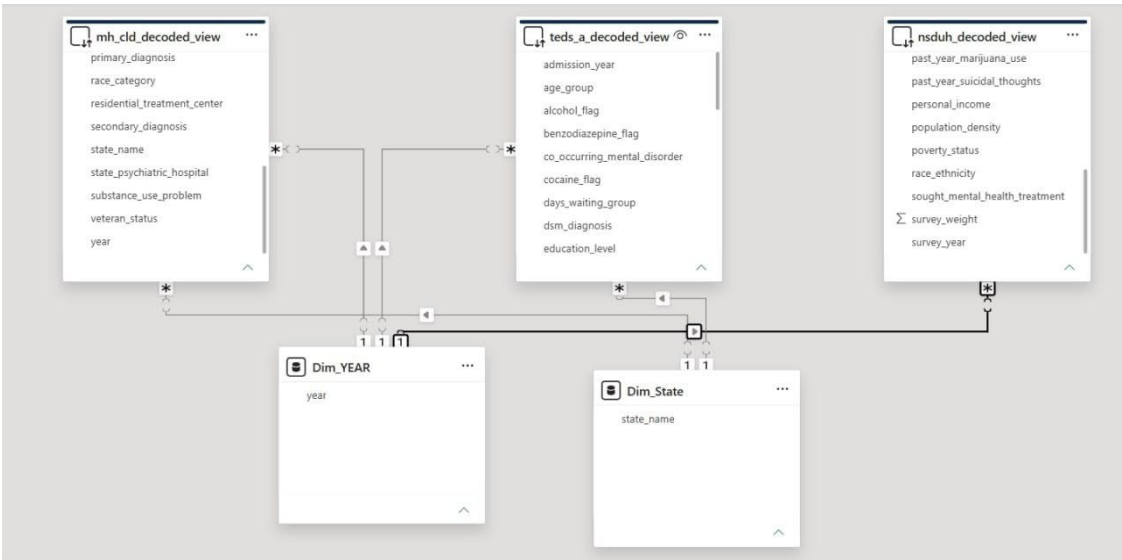


Figure 1 — Power BI data model: Star Schema relating MH-CLD, TEDS-A, and NSDUH decoded views through `Dim_YEAR` and `Dim_State`.

Logical & Physical Schema

Table / View	Key Attributes
<code>mh_cld_decoded_view</code>	<code>primary_diagnosis</code> , <code>race_category</code> , <code>residential_treatment_center</code> , <code>secondary_diagnosis</code> , <code>state_name</code> , <code>state_psychiatric_hospital</code> , <code>substance_use_problem</code> , <code>veteran_status</code> , <code>year</code>
<code>teds_a_decoded_view</code>	<code>admission_year</code> , <code>age_group</code> , <code>alcohol_flag</code> , <code>benzodiazepine_flag</code> , <code>co_occurring_mental_disorder</code> , <code>cocaine_flag</code> , <code>days_waiting_group</code> , <code>dsm_diagnosis</code> , <code>education_level</code>
<code>nsduh_decoded_view</code>	<code>past_year_marijuana_use</code> , <code>past_year_suicidal_thoughts</code> , <code>personal_income</code> , <code>population_density</code> , <code>poverty_status</code> , <code>race_ethnicity</code> , <code>sought_mental_health_treatment</code> , <code>survey_weight</code> (measure), <code>survey_year</code>
<code>Dim_YEAR</code>	<code>year</code> (shared conformed dimension across all three datasets)
<code>Dim_State</code>	<code>state_name</code> (shared conformed dimension for MH-CLD and TEDS-A)

Normalization note: state_name and year are decoupled into conformed dimensions so the three source datasets share consistent filtering without being joined to one another directly — satisfying the "0 direct many-to-many relations" KPI.

3.3 Data Flow & System Behavior

DFD (Data Flow Diagram)

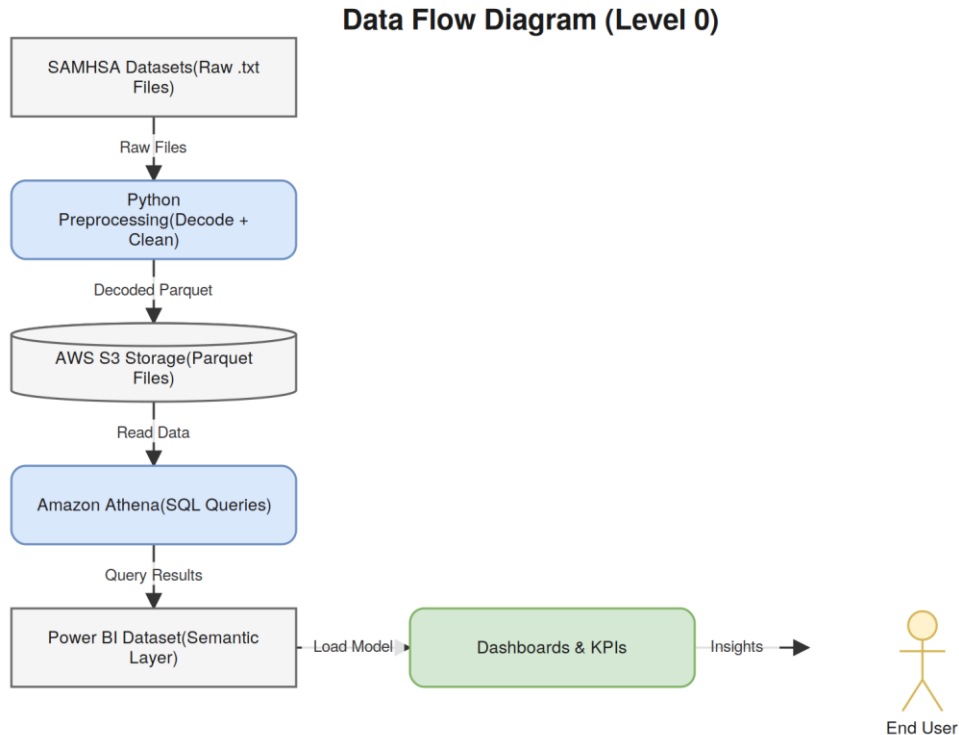


Figure 3 — Data Flow Diagram (Level 0): raw SAMHSA files are decoded and compressed to Parquet, staged in AWS S3, queried through Amazon Athena, and surfaced to end users through the Power BI semantic layer.

Sequence Diagrams

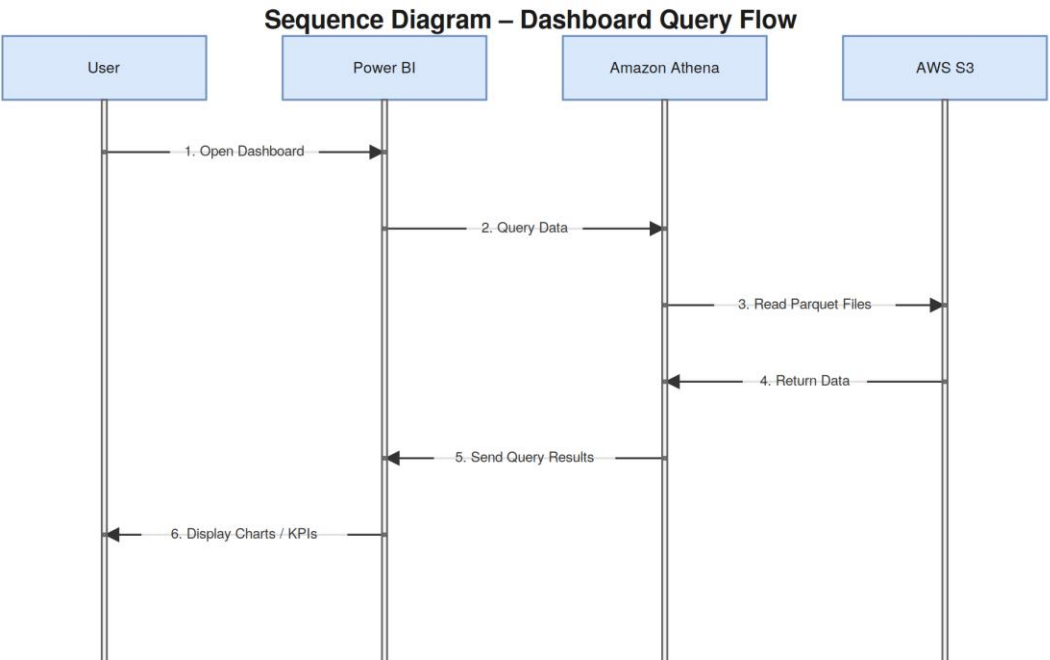


Figure 4 — Sequence Diagram: message flow when a user opens the dashboard, from the Power BI query through Amazon Athena and AWS S3 and back to the rendered charts and KPIs.

Activity Diagram

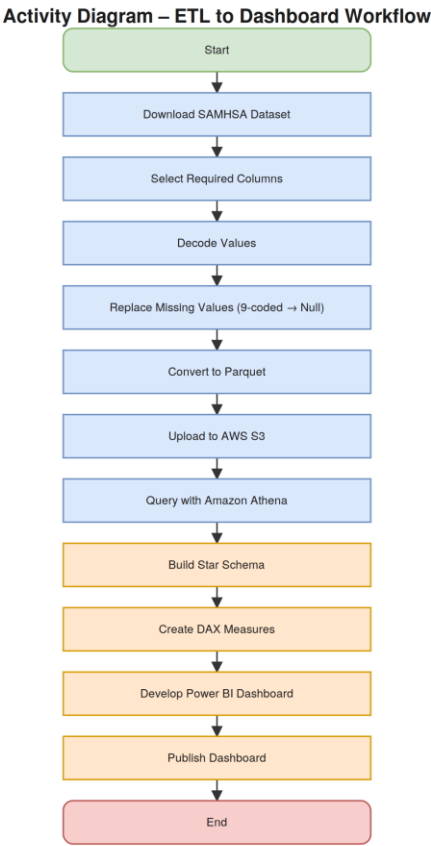


Figure 5 — Activity Diagram: end-to-end workflow from dataset download through preprocessing, star-schema modeling, and dashboard publication.

State Diagram

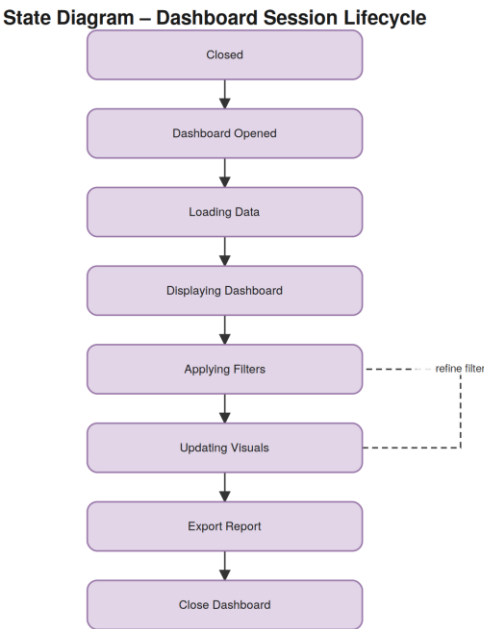


Figure 6 — State Diagram: lifecycle of a dashboard session, from being closed to loading data, displaying visuals, applying filters, and exporting reports.

Class Diagram

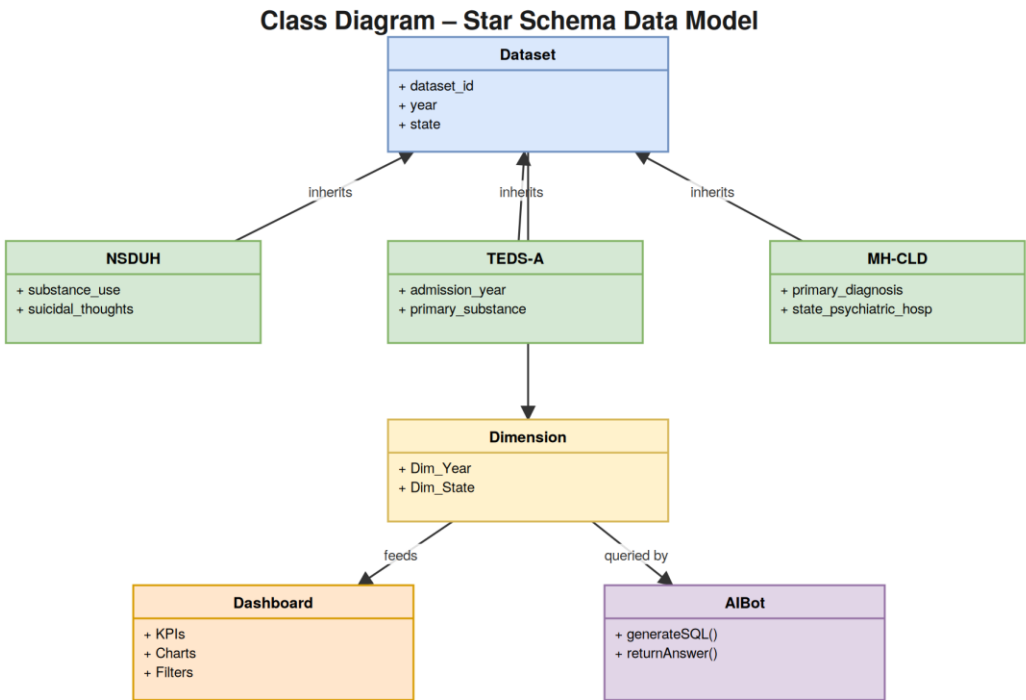


Figure 7 — Class Diagram: NSDUH, TEDS-A, and MH-CLD each extend a common Dataset class and connect through the shared Dimension (Year, State) class to the Dashboard and AIBot classes.

3.4 UI/UX Design & Prototyping

Wireframes & Mockups

The project delivers three linked Power BI dashboards — one per SAMHSA dataset — sharing a consistent dark-theme navigation shell, KPI-card layout, and filter pattern so users can move between them without relearning the interface.

TEDS-A Dashboard — Overview Page



Figure 8 — TEDS-A (Treatment Episode Data Set: Admissions) Overview page.

- Left navigation rail: Overview, Patients, Drugs & Treatment, Geographic & Comparative, Recommendations.
- Top slicer bar: year, gender, education level, state name, primary substance.
- KPI cards: most-used substance (Alcohol), total primary substances (20), total cases (4.74M), total routes of administration (6).
- Horizontal bar charts for patients by education level and by marital status, with values labelled directly on the bars.

MH-CLD Dashboard — Overview Page



- Left navigation rail: Overview, Demographic, Profile, Diagnosis, PD vs SD, Services, Map, Recommendations.
- Top slicer bar: year, state name, gender, primary diagnosis.
- KPI cards: total patients (21M), total states (52), total primary diagnoses (14), and a gender-distribution donut chart (Male 47% / Female 53%).
- Line chart of total cases by year, a ranked bar chart of the top 5 primary diagnoses, a horizontal bar chart of the top 5 states by case volume, and an age-group distribution column chart.

NSDUH Dashboard — Overview Page



- Left navigation rail: Overview, Mental Health, Substance Use, Psychological Distress, Treatment Gap, Recommendations.
- Top slicer bar: survey year, age category, gender.

- KPI cards: total respondents (174K), alcohol-use rate (62.50%), marijuana-use rate (20.96%), cigarette-use rate (18.07%).
- A dual-line chart comparing total respondents against estimated population by survey year, plus grouped bar charts of substance-use rate by depression status and by suicide ideation status.

UI/UX Guidelines

Design Principles

- Consistency — all three dashboards reuse the same shell: a fixed left navigation rail, a top slicer bar, and a KPI-card row directly under it, so users transfer knowledge between NSDUH, TEDS-A, and MH-CLD without relearning the layout.
- Progressive disclosure — each dashboard opens on an Overview page (headline KPIs and trend charts) and pushes deeper breakdowns (demographics, diagnosis, geography, services) to dedicated pages reached from the navigation rail.
- Direct manipulation — every chart is filterable through the shared slicer bar and supports cross-filtering, so selecting a bar or category updates the KPIs and other visuals on the same page instantly.
- Data-ink focus — values are labelled directly on bars and cards rather than relying on separate legends, keeping the reader's attention on the numbers.

Color Scheme

- Base — a dark navy background (approx. #0B1F3A–#13294B) with a subtle network/hexagon watermark pattern, chosen to reduce eye strain during long analysis sessions and give the dashboards a clinical, data-driven feel.
- Foreground — white and light-gray text and chart elements (approx. #FFFFFF / #D9D9D9) for high contrast against the dark base.
- Accent — a single muted slate-blue tone is used consistently for bars, KPI icons, and highlighted values, avoiding a multi-color palette that would compete with the data.
- Navigation — the active page button is filled solid while inactive buttons stay outlined, giving a clear, low-noise wayfinding cue.

Typography

- A single sans-serif typeface is used throughout for legibility on screen.
- KPI values are set large and bold (the dominant visual weight on each page) with a smaller, lighter caption underneath (e.g., "174K" over "Total Respondents"), so the headline number reads before the label.
- Chart and slicer titles use a smaller, medium-weight style to stay subordinate to the KPI cards.

Accessibility Considerations

- Contrast — white-on-navy text meets WCAG AA contrast for normal text; this should be spot-checked for the lighter gray chart labels specifically.
- Color independence — because the accent palette is largely single-hue, category differences rely on bar position/labels rather than color alone, which helps color-blind users; any remaining multi-series charts (e.g., substance-use-rate comparisons) should keep direct data labels for the same reason.
- Filter feedback — slicers should always show the current selection (e.g., "All" vs a chosen year) so screen-reader and keyboard users can confirm dashboard state without relying on visual context alone.
- Navigation — the left-rail buttons are large, clearly labelled targets suitable for touch and keyboard navigation alike.

3.5 AI Bot Design

The AI Bot is the Text-to-SQL Q&A layer referenced in the Use Case Diagram, DFD, Sequence Diagram, and Class Diagram above (Section 3.1–3.3), and in the Deployment and Component Diagrams in Section 4. It gives non-technical stakeholders — clinicians, policy analysts, and healthcare executives — a conversational,

plain-English alternative to writing SQL against the Athena/S3 data lake. This subsection documents its design at the architecture, workflow, and message-flow level.

Architecture

The bot is organized into four layers, all built on the same cloud data lake used by the Power BI dashboards (Section 3.1):

- **Presentation layer** – a Streamlit chat and insights interface where the user types questions and views charts.
- **AI reasoning layer** – Google Gemini, used for intent classification, SQL generation, result summarization, and insight narration.
- **Query execution layer** – Amazon Athena, executing the generated SQL against the same Parquet-compressed views on AWS S3 that feed the Power BI Star Schema.
- **Audit layer** – a local, password-gated log recording every AI-generated query for compliance review, in line with the "who ran what query" requirement in Section 2's stakeholder analysis.

Workflow

1. User asks a plain-English question in the chat interface.
2. An intent-classification step routes the question as schema-only or data-requiring.
3. For data questions, an Athena-safe SQL query is generated, grounded in a cached copy of the schema so no extra Athena call is needed just to look up column names.
4. The query runs against Amazon Athena / AWS S3, with a timeout so a slow query cannot hang the interface.
5. Results are summarized in clinical language and rendered as an interactive chart, styled to match the Power BI theme in Section 3.4.
6. The question, generated SQL, and result metadata are written to the audit log.

Sequence Diagram

This extends the dashboard-open sequence in Figure 4 (Section 3.3) with the bot's own message flow: User → Chat UI → Intent Classifier; on a data question, Chat UI → SQL Generator → Amazon Athena → AWS S3 → Athena (rows) → Chat UI, which converts the result to a chart and a written summary and returns both to the user. In parallel, the Chat UI writes the question, SQL, and outcome to the audit log after each turn.

Component Diagram

As shown at the system level in Figure 12 (Section 4), the AI Bot sits alongside the Power BI Dashboard as a second consumer of the same Athena/S3 layer. Internally, it breaks down into four components: a Chat & Insights UI, an AI Reasoning Engine (intent classifier, SQL generator, summarizer, and insight planner), a Query Execution component wrapping the Athena client, and an Audit Store for the query log.

Prompt Flow

A cached copy of the schema is injected into every prompt sent to Gemini, so the model always has grounding context without an extra Athena call. From there, prompts chain in stages:

- **Classification prompt** – schema + question, routed to a schema-only answer or a SQL-generation path.
- **SQL-generation prompt** – schema + question, producing a single Athena-safe query.
- **Summarization prompt** – question + result rows, producing a short clinical-language narrative.
- **Insight-plan prompt** – for guided exploration, returns a structured chart plan (title, SQL, chart type, axes), followed by a narration prompt for the accompanying takeaway.

4. System Deployment & Integration

Technology Stack

- **Storage** – AWS S3 with Parquet-compressed datasets.
- **Query Engine** – Amazon Athena.
- **BI / Presentation** – Power BI (Star Schema, DAX measures, Semantic Layer).
- **AI Layer** – a Streamlit chat front-end calling Google Gemini for natural-language reasoning (intent classification, SQL generation, summarization), which in turn executes the generated SQL through the same Amazon Athena / AWS S3 layer used by Power BI.

In short, Streamlit is the AI Bot's presentation layer, Google Gemini is its reasoning engine, and Amazon Athena / AWS S3 is the shared query engine and data lake it queries alongside Power BI — see Section 3.5 for the bot's internal design.

Deployment Diagram

Deployment Diagram – Cloud Infrastructure

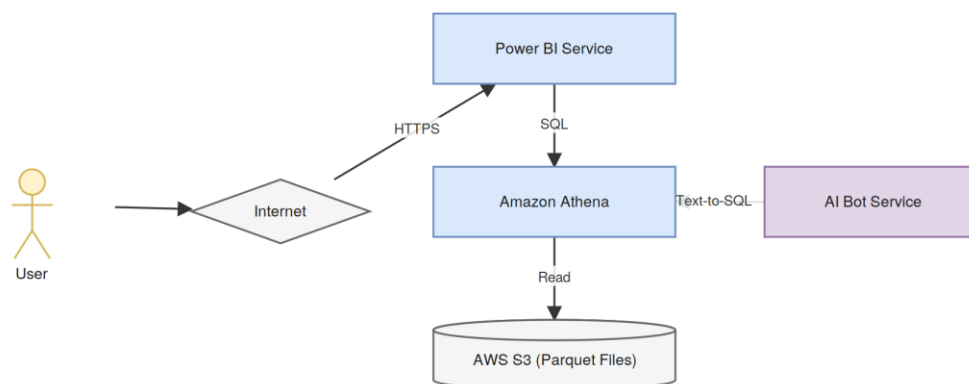


Figure 11 — Deployment Diagram: the user reaches the Power BI Service over the internet; Power BI and the AI Bot both query Amazon Athena, which reads Parquet files from AWS S3.

Component Diagram

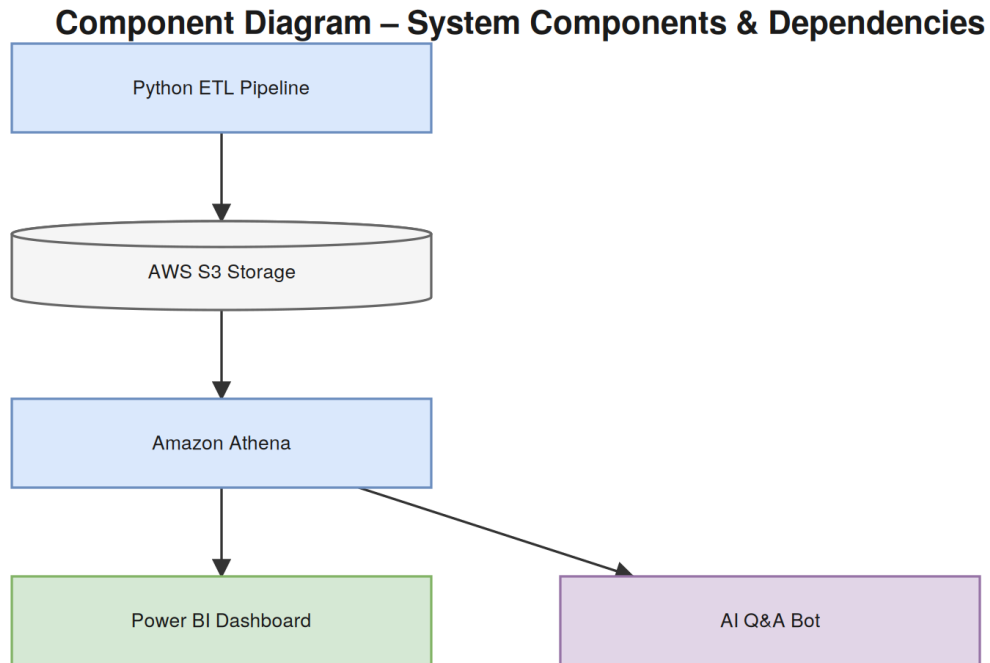


Figure 12 — Component Diagram: the Python ETL pipeline loads AWS S3, which is queried by Amazon Athena and consumed by both the Power BI Dashboard and the AI Q&A Bot.

5. Testing & Quality Assurance

Per Phase 6 (Review & Documentation, Jul 2 – Jul 8, 2026) of the project plan, quality assurance is carried out through:

- **Team Quality Review** – the full team reviews preprocessing scripts, the star-schema model, and dashboard outputs before submission.
- **Final Data Validation** – the three decoded datasets (MH-CLD, TEDS-A, NSDUH) are validated against the star schema and KPI targets (e.g., 100% conversion of 9-coded flags to Null, 0 direct many-to-many relations) prior to final delivery.
- **Delivery Tracking** – progress is measured against KPI 7 (6 of 7 milestones completed on time) and KPI 8 (final submission by July 10, 2026).

Validation Performed

Validation is carried out at four levels, mapped to the KPIs defined in Section 1.5 and the model documented in Section 3.2:

- **Data validation** – confirming that 9-coded flags are converted to Null across all three decoded views (KPI 2), and spot-checking row counts and key aggregate totals against the raw SAMHSA source files for each dataset year.
- **Model validation** – confirming that Dim_YEAR and Dim_State enforce the "0 direct many-to-many relations" target (KPI 4) by testing filter and slicer interactions across all three dashboards rather than relying on a single sample page.
- **Dashboard validation** – a manual walkthrough of every page on all three dashboards, confirming that KPI cards, slicers, and cross-filtering update consistently and that navigation between pages behaves as designed in Section 3.4.
- **AI Bot validation** – reviewing a sample of Gemini-generated SQL queries recorded in the audit log (Section 3.5) for correctness before they are trusted, and confirming that the intent classifier correctly routes schema-only questions away from full SQL generation.

Together with the Team Quality Review and Final Data Validation above, this validation pass is tracked against KPI 7 and KPI 8 ahead of final submission.

6. Final Presentation & Reports

Technical Documentation

System architecture, database schema, AI Bot design, and deployment integration are covered in Sections 3 and 4 above.

Project Presentation (PPT/PDF)

Final presentation is scheduled for July 17, 2026, following Phase 7 (Jul 8 – Jul 17, 2026): deck preparation, team rehearsal, and dashboard walkthrough.

Presentation Link: [View the Project Presentation \(Google Slides / PPT\)](#)

Conclusion

This project delivered an end-to-end analytics pipeline for SAMHSA mental health data — from raw ingestion through Parquet-optimized storage, a Star Schema semantic model, and interactive Power BI dashboards, backed by an AI Text-to-SQL bot for natural-language exploration. Across the NSDUH, TEDS-A, and MH-CLD datasets, the team met its KPI-driven targets for storage reduction, data quality, and modeling integrity, translating fragmented public datasets into decision-ready insight for healthcare policymakers, analysts, and clinicians.

- **High-Quality Mental Health Analytics** – decoded, validated views across three SAMHSA datasets supporting reliable demographic and treatment-trend analysis.
- **Optimized Data Engineering Pipeline** – Parquet compression and a Star Schema on AWS S3/Amazon Athena, built for performance and scalability.
- **Interactive Power BI Delivery** – three linked dashboards with a semantic layer and an AI Q&A bot, giving stakeholders self-service access to the findings.

Final Presentation: July 17, 2026